

Research paper report and presentation

STA2053

Overview

The goal of this assessment is for you to read and understand a paper where Bayesian hierarchical models have been used to solve a problem in an applied setting. When reading the paper, I want you to not only understand the advantages of using the model the authors chose, but read with a critical eye and think about any limitations, or alternative analysis choices that could have been made.

The papers

Please choose one paper from the following:

- Alkema, L., Raftery, A. E., Gerland, P., Clark, S. J., Pelletier, F., Buettner, T., & Heilig, G. K. (2011). Probabilistic projections of the total fertility rate for all countries. *Demography*, 48(3), 815-839.
- Cahill, N., Kemp, A. C., Horton, B. P., & Parnell, A. C. (2016). A Bayesian hierarchical model for reconstructing relative sea level: from raw data to rates of change. *Climate of the Past*, 12(2), 525-542.
- Lin, J., Myers, M. F., Koehly, L. M., & Marcum, C. S. (2019). A Bayesian hierarchical logistic regression model of multiple informant family health histories. *BMC Medical Research Methodology*, 19(1), 56.
- Virgili-Gervais, G., Schmidt, A. M., Bixby, H., Cavanaugh, A., Owusu, G., Agyei-Mensah, S., ... & Baumgartner, J. (2025). Mapping socio-economic status using mixed data: a hierarchical Bayesian approach. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 188(3), 859-874.
- Wang, W., Rothschild, D., Goel, S., & Gelman, A. (2015). Forecasting elections with non-representative polls. *International journal of forecasting*, 31(3), 980-991.

Report (10%)

Write a summary report of your chosen paper. The report should be written in Quarto or R Markdown, and follow a logical format. It should include the following information:

- Background/motivation of the paper
- Outline of statistical model
- Data use and summary of results
- Strengths and limitations of the paper. Think about: why was a Bayesian hierarchical model chosen? Was it useful/justified given the context? What other modeling approaches could have been considered? Are there obvious limitations? Any possible extensions?

This will be due in Week 12.

Presentation (15%)

In Week 12 you will give a short presentation on your paper, summarizing your report. You should prepare slides, and aim for around 10-15 mins.